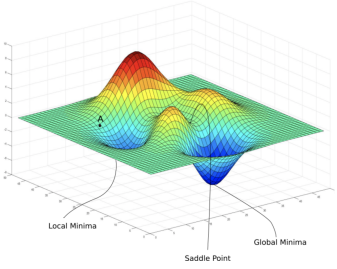
# Optimization in Machine Learning

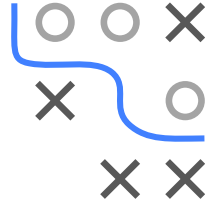
## Foundations
## Conditions for optimality




**Learning goals**

- Local and global optima
- Strict, isolated, and unique minimizers
- First & second order conditions

# EXTREMA AND SADDLE POINTS



- Given $\mathcal{S} \subseteq \mathbb{R}^d$, $f : \mathcal{S} \to \mathbb{R}$
- Global minimum at $\boldsymbol{x}^*$: $f(\boldsymbol{x}^*) \leq f(\boldsymbol{x})$ for all $\boldsymbol{x} \in \mathcal{S}$
- Local minimum at $\boldsymbol{x}^*$: $\exists\, \epsilon > 0$ s.t. $f(\boldsymbol{x}^*) \leq f(\boldsymbol{x})$ for all $\boldsymbol{x} \in \mathcal{S} \cap B_\epsilon(\boldsymbol{x}^*)$ ($\epsilon$-ball)
- Analogously for global and local max
- We call $\boldsymbol{x}^*$ saddle point if in feasible portion of every eps-ball $\mathcal{S} \cap B_\epsilon(\boldsymbol{x}^*)$, is at least a strictly better and a strictly worse point

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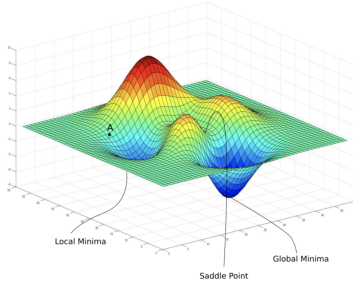


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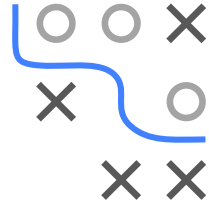
# VARIANTS OF MINIMIZERS



- Several refined notions for $\boldsymbol{x}^* \in \mathcal{S}$:

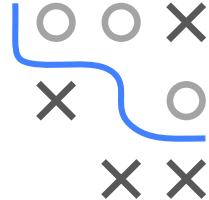
**cal min:** $f(\boldsymbol{x}^*) < f(\boldsymbol{x})$ for all $\boldsymbol{x} \in \mathcal{S} \cap B_\epsilon(\boldsymbol{x}^*)$ with $\boldsymbol{x} \neq \boldsymbol{x}^*$

**cal min:** $\boldsymbol{x}^*$ is a local min and $B_\epsilon(\boldsymbol{x}^*)$ contains no other local min

**bal min:** $\boldsymbol{x}^*$ is the only global minimizer on $\mathcal{S}$

- Hierarchy: unique global $\Rightarrow$ isolated local $\Rightarrow$ strict local $\Rightarrow$ local
- Reverse implications generally fail
- **Caveat:** strict local does NOT imply isolated – example $f(x) = x^4(2 + \sin(1/x))$ at $x = 0$

# EXISTENCE OF OPTIMA

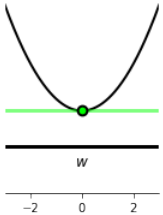


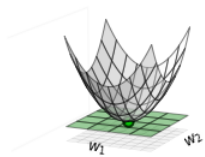
- $f : \mathcal{S} \to \mathbb{R}$
- If $f$ continuous and $\mathcal{S}$ compact: minimum and maximum exist (extreme value theorem)
- If $f$ discontinuous: no general existence statement
- Negative example, with $\mathcal{S} = [0, 1]$:

$$f(x) = \begin{cases} 1/x & x > 0 \\ 0 & x = 0 \\ -1/x & x < 0 \end{cases}$$

# FIRST-ORDER CONDITION

- Let $f : \mathcal{S} \to \mathbb{R}$, $f$ differentiable, $\boldsymbol{x}^*$ interior point of $\mathcal{S}$
- Necessary condition:
  If $\boldsymbol{x}^*$ is a local extremum, then $\nabla f(\boldsymbol{x}^*) = 0$
- Such points are called "stationary"
- Intuition: at a local extremum, the function must be flat, otherwise we can find a direction to move to a better value
- Not sufficient, e.g. saddle points are possible

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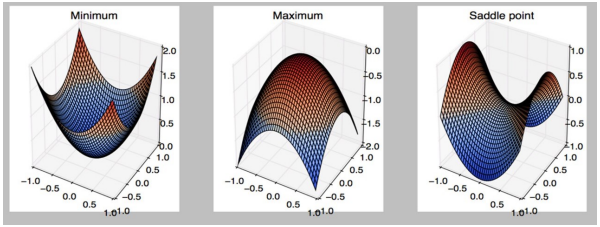
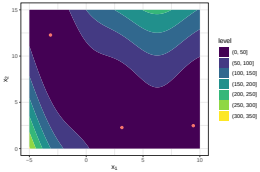
# SECOND-ORDER CONDITION

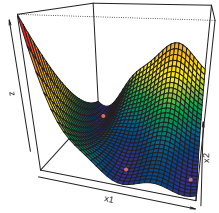
- Let $f : \mathcal{S} \to \mathbb{R}$, $f \in \mathcal{C}^2$, $\boldsymbol{x}^*$ interior point of $\mathcal{S}$
- If $H(\boldsymbol{x}^*)$ is definite, then $\boldsymbol{x}^*$ is a strict local extremum
- If $H(\boldsymbol{x}^*)$ is semi-definite, then $\boldsymbol{x}^*$ is a local extremum
- If $H(\boldsymbol{x}^*)$ is indefinite, then $\boldsymbol{x}^*$ is a saddle point
- If $H(\boldsymbol{x}^*)$ is p(s)d, then $\boldsymbol{x}^*$ is a (strict) local min
  this implies $f$ is locally (strictly) convex
- If $H(\boldsymbol{x}^*)$ is n(s)d, then $\boldsymbol{x}^*$ is a (strict) local max
  this implies $f$ is locally (strictly) concave
- Interpretation: curvature pos (or neg) in all directions



# EXAMPLE: BRANIN FUNCTION

- Branin function with 3 local minima

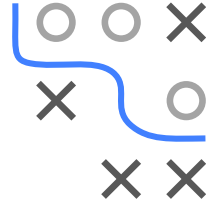



- EVs of Hessian at local minima:

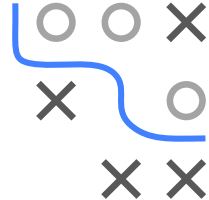
| | $\lambda_1$ | $\lambda_2$ |
|---|---|---|
| Left | 22.29 | 0.96 |
| Middle | 11.07 | 1.73 |
| Right | 11.33 | 1.69 |

# CONVEXITY AND OPTIMA



- $f : \mathcal{S} \to \mathbb{R}$ convex on convex set $\mathcal{S}$
- Any local minimum is global
- The set of minima is convex
- If $f$ strictly convex: at most one local minimum (unique global on $\mathcal{S}$, if it exists)
- Analogously for concave functions

# CONVEXITY AND OPTIMA: PROOFS



**l = global.** Assume $\boldsymbol{x}^*$ local min but not global, so $\exists \bar{\boldsymbol{x}} \in \mathcal{S}$ with $f(\bar{\boldsymbol{x}}) < f(\boldsymbol{x}^*)$. By convexity, $\forall t \in [0, 1]$:

$$f(\boldsymbol{x}^* + t(\bar{\boldsymbol{x}} - \boldsymbol{x}^*)) \leq (1 - t)f(\boldsymbol{x}^*) + tf(\bar{\boldsymbol{x}}) < f(\boldsymbol{x}^*)$$

For $t$ small enough, the point $\boldsymbol{x}^* + t(\bar{\boldsymbol{x}} - \boldsymbol{x}^*) \in B_\epsilon(\boldsymbol{x}^*) \cap \mathcal{S}$ – contradiction.

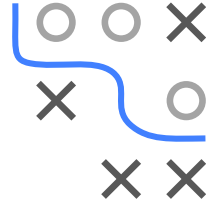
**na convex.** Let $\boldsymbol{x}^*, \bar{\boldsymbol{x}}$ both be global mins, so $f(\boldsymbol{x}^*) = f(\bar{\boldsymbol{x}}) = f^*$. By convexity, $\forall t \in [0, 1]$:

$$f(\boldsymbol{x}^* + t(\bar{\boldsymbol{x}} - \boldsymbol{x}^*)) \leq (1 - t)f^* + tf^* = f^*$$

Since $f^*$ is the global min, equality must hold, so $\boldsymbol{x}^* + t(\bar{\boldsymbol{x}} - \boldsymbol{x}^*)$ is also a global minimizer.

**t one min.** Same setup with $\boldsymbol{x}^* \neq \bar{\boldsymbol{x}}$, both global mins. Strict convexity gives **strict** inequality $f(\boldsymbol{x}^* + t(\bar{\boldsymbol{x}} - \boldsymbol{x}^*)) < f^*$ for $t \in (0, 1)$ – contradiction.

# EXAMPLE



- $f(x, y) = x^4 + y^4 - x^2 - y^2$
- $\nabla f(x, y) = (4x^3 - 2x, 4y^3 - 2y)$
- $H(x, y) = \begin{pmatrix} 12x^2 - 2 & 0 \\ 0 & 12y^2 - 2 \end{pmatrix}$
- At $(0, 0)^T$ we have strict local max
- At $(\pm\frac{1}{\sqrt{2}}, \pm\frac{1}{\sqrt{2}})^T$ we have 4 strict local min
- At $(0, \pm\frac{1}{\sqrt{2}})^T$, $(\pm\frac{1}{\sqrt{2}}, 0)^T$ we have 4 saddle points